Emelia May Hughes

PhD Candidate in Computer Science and Engineering, University of Notre Dame

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Education

University of Notre Dame

Ph.D. Candidate, Computer Science and Engineering

Aug 2023 - present

- Advisor: Dr. Tim Weninger
- Remick Graduate Fellow
- Expected May 2028

University of Washington

B.S. Informatics and B.A. Art

Aug 2018 - Jun 2022

- Bachelor of Science in Informatics, Human-Computer Interaction
- Bachelor of Arts with Honors in Art, Drawing and Painting

Research Experience

Weninger Lab, University of Notre Dame

Graduate Researcher

Aug 2023 - present

- Advisor: Dr. Tim Weninger
- Research how algorithmically mediated platforms shape online community formation, visibility, belonging, and public discourse
- Develop dissertation research on feed-based platforms, social inference, and emergent belonging using qualitative, survey, and computational methods
- Study platform communities and information systems including TikTok and Truth Social

Social Futures Lab, University of Washington

Research Assistant

Aug 2020 - Aug 2023

- Advisor: Dr. Amy Zhang
- Led a WikiCred-funded project developing new approaches to credibility and misinformation on YouTube
- Conducted contextual-inquiry interviews about user needs surrounding credibility
- Designed and developed the Viblio browser extension and companion application using Adobe XD and React

Center for an Informed Public, University of Washington

Undergraduate Researcher

Jan 2021 - Jun 2022

- Advisor: Dr. Rachel Moran
- Analyzed the #SaveTheChildren movement on Instagram and its relationship to QAnon
- Developed a qualitative codebook used to analyze more than 1,400 social media posts

Adobe Research

Research Intern

Jun 2021 - Sep 2021

- Mentor: Dr. Jose Echevarria
- Designed an accessible experience for artists with visual impairments to navigate canvas-based applications
- Conducted background research, communicated findings to company stakeholders, and implemented an iPad prototype in Swift

Publications

Peer-Reviewed Publications

1. **Emelia May Hughes**, Renee Wang, Prerna Juneja, Tony W. Li, Tanushree Mitra, and Amy X. Zhang. 2024. Viblio: Introducing Credibility Signals and Citations to Video-Sharing Platforms. Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24), Article 807, 1-20. https://doi.org/10.1145/3613904.3642490

2. Moran, R. E., Prochaska, S., Schlegel, I., **Hughes, E. M.**, & Prout, O. (2021). Misinformation or Activism: Mapping Networked Moral Panic Through an Analysis of #SaveTheChildren. AoIR Selected Papers of Internet Research. https://doi.org/10.5210/spir.v2021i0.12212 Extended abstract.

Posters, Companion Papers, and Symposium Presentations

1. **Hughes, E. M.**, & Weninger, T. (2025). Echo Chambers and Information Brokers on Truth Social: A Study of Network Dynamics and Political Discourse. CSCW Companion '25. https://doi.org/10.1145/3715070.3749241 Poster.

2. Oguine, O. C., Anuyah, O., **Hughes, E. M.**, & Badillo-Urquiola, K. (2024). Examining Mainstream News Media Narratives on Youth Online Safety. CSCW Companion '24, 349-354. https://doi.org/10.1145/3678884.3681874

3. **Hughes, E. M.**, Wang, R., Huang, A., Juneja, P., Mitra, T., & Zhang, A. X. (2022). Introducing Credibility Signals and Citations to Video-Sharing Platforms. University of Washington Undergraduate Research Symposium.

4. **Hughes, E. M.** (2021). WikiUX: Introducing Credibility Signals and Citations to YouTube. University of Washington Undergraduate Research Symposium.

Preprints

1. **Hughes, E.**, & Weninger, T. (2026). Ephemeral Feeds and Enduring Rituals: RushTok and the Formation of Event-Based Algorithmic Communities. arXiv preprint. https://arxiv.org/abs/2609.09331

2. **Hughes, E.**, & Weninger, T. (2026). Echoes in the Algorithm: Analyzing the Fidelity of User Preferences Against Realized Platform Reach. arXiv preprint. https://arxiv.org/abs/2609.09365

Doctoral Consortium and Workshop Papers

1. **Hughes, E. M.** (2024). #RushTok: An Evaluation of Event-Based Communities on TikTok. Association of Internet Researchers Doctoral Colloquium.

2. **Hughes, E. M.**, Wang, R., Juneja, P., Mitra, T., & Zhang, A. X. (2021). Introducing Credibility Signals and Citations to Video-Sharing Platforms. ICWSM Workshop on Information Credibility & Alternative Realities in Troubled Democracies.

Patents

1. “Multimodal canvas landmarks for accessible digital painting.” J. Echevarria, J. Sun, and **E. Hughes**. Filed 2023.

Fellowships and Awards

2026

- Notre Dame-IBM Technology Ethics Fellow

2025

- Graduate Justice Fellowship, Institute for Social Concerns, University of Notre Dame

2024

- Doctoral Colloquium, Association of Internet Researchers
- Notre Dame Computer Science and Engineering Poster Fair, Best Research Pitch, Honorable Mention

2023

- Remick Graduate Fellowship in Engineering, University of Notre Dame

2022

- RoboMaster North American Champions (3v3 and 1v1)
- Women United Art Prize, Shortlisted Artist
- Priscilla and Melvin Wilson Scholarship, University of Washington School of Art

2021

- RoboMaster North American Champions (3v3)
- Adobe Women in Technology Finalist

2020

- RoboMaster Second Prize, Open Source Award

2019

- RoboMaster, Shenzhen, China: Second Prize, International Regionals; Outstanding PR Manager Award; Third Prize, Hero Remote Firing; Most Aesthetic Robots; Outstanding Award A Level, Open Source Award

Academic and Professional Service

Conference Reviewer

CSCW, CHI, and ICWSM

2025 - 2026

- CSCW 2025; CHI 2026; ICWSM 2025 and 2026

Notre Dame Graduate Society of Women Engineers

Executive Board Secretary

Spring 2026

- Maintain meeting records and organizational documentation while supporting event logistics and programming
- Produce the newsletter and support member communications around professional development, mentorship, and community engagement

501(c)(3) Supporting Advanced Robotics at the University of Washington

President of the Board

2023 - present

- Provide nonprofit governance and organizational support for ARUW

Advanced Robotics at the University of Washington

President

2021 - 2022

- Led a 75-person student robotics team to the 2022 North American RoboMaster Championship
- Coordinated team operations, scheduling, competition logistics, and organizational decision-making
- Redesigned recruitment and application processes to broaden participation and improve equity

Advanced Robotics at the University of Washington

Vice President

2019 - 2021

- Supported project management, sponsorship, and coordination across design, build, and competition cycles

Teaching Experience

Current State and Future of Generative AI (ESTM 60244)

Instructor of Record, University of Notre Dame ESTEEM Master's Program

Fall 2025; Fall 2026

- Designed and developed a graduate course on generative AI, prompt design, ethics, and AI product development
- Created course materials, assignments, readings, discussions, and project structure
- Led class discussions on AI ethics, bias, labor, and responsible use of generative systems

Elements of Computing I (CSE 10101)

Instructor of Record, University of Notre Dame

Spring 2026

- Taught introductory programming and computational thinking to non-major undergraduate students
- Designed instruction, assignments, and assessments for students with varied technical backgrounds

Human Computer Interaction (CSE 40424)

Teaching Assistant, University of Notre Dame

Spring 2025

- Instructor: Annalisa Szymanski
- Supported grading, office hours, assignment design, and student project mentorship

Database Concepts (CSE 30246)

Teaching Assistant, University of Notre Dame

Fall 2023; Fall 2024

- Instructor: Dr. Tim Weninger
- Supported grading, office hours, and individual student support across two semesters

Forest Ridge School

Long-Term Substitute Teacher

Mar 2023 - Jun 2023

- Taught Calculus II and designed a new Web Development curriculum after taking over both courses mid-year

St. Catherine of Siena School

Robotics Teacher

Sep 2022 - Jun 2023

- Designed and taught a middle-school robotics elective using LEGO Mindstorms

St. Catherine of Siena School

Extended Care Program Assistant

Sep 2022 - Jun 2023

- Worked alongside the program director to run the after-school care program for kindergarten through eighth grade

Private & Grade Potential Tutoring

Tutor

2016 - 2023

- Developed individual learning plans in mathematics (algebra through calculus), physics, chemistry, SAT/ACT preparation, English, and writing

Outreach and Community Engagement

2025

- Event volunteer and livestream host, RoboMaster North America Competition, San Diego

2024

- Event volunteer and livestream host, RoboMaster North America Competition, Boulder

2021 - 2022

- Student volunteer, ACM CHI 2021
- Invited speaker, Humboldt State University Robotics Camp
- ARUW volunteer and speaker for admitted-student, engineering orientation, and recruitment events

2019 - 2021

- Organizer and volunteer for ARUW Dawg Daze, Engineering Discovery Days, Giving Day, and STEM outreach
- Volunteer English streaming host, RoboMaster 2019

2019 - 2022

- ARUW: Admitted Student Preview volunteer (2022); Mechanical Engineering and Electrical Engineering orientation speaker and volunteer (2020, 2022); School of Engineering Kick-off volunteer (2020-2022)
- ARUW: Dawg Daze organizer (2019-2021); Giving Day and Engineering Discovery Days volunteer (2019); Sammamish High School STEM Fair volunteer (2019)